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A BAYESIAN APPROACH TO CLASSIFICATION CRITERIA FOR SPECTACLED EIDERS[1,2]

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Abstract. To facilitate decisions to classify species according to risk of extinction, we used Bayesian methods to analyze trend data for the Spectacled Eider, an arctic sea duck. Trend data from three independent surveys of the Yukon–Kuskokwim Delta were analyzed individually and in combination to yield posterior distributions for population growth rates. We used classification criteria developed by the recovery team for Spectacled Eiders that seek to equalize errors of under- or overprotecting the species. We conducted both a Bayesian decision analysis and a frequentist (classical statistical inference) decision analysis. Bayesian decision analyses are computationally easier, yield basically the same results, and yield results that are easier to explain to nonscientists. With the exception of the aerial survey analysis of the 10 most recent years, both Bayesian and frequentist methods indicated that an endangered classification is warranted. The discrepancy between surveys warrants further research. Although the trend data are abundance indices, we used a preliminary estimate of absolute abundance to demonstrate how to calculate extinction distributions using the joint probability distributions for population growth rate and variance in growth rate generated by the Bayesian analysis. Recent apparent increases in abundance highlight the need for models that apply to declining and then recovering species.

Key words: Bayesian inference; decision analysis; extinction; risk classification; Spectacled Eider; trends; uncertainty.

INTRODUCTION

Organizations charged with protecting vulnerable species face dilemmas in deciding which species are most in need of their limited resources and, at least in many public agencies, whether or not to protect a species at the expense of other social, environmental, or economic objectives. To assist in the allocation of protection actions, vulnerable species are often classified according to their risk of extinction. For example, in the United States, the Endangered Species Act of 1973, as amended, formalizes two risk categories: endangered (in danger of extinction) and threatened (likely to become endangered within the foreseeable future). Yet, classification of species has been problematic due to limited data and a lack of objective procedures to assist decision making based on uncertain data. We present a new approach using a Bayesian analysis of population trend data to classify species, with a framework that was developed by the recovery team for the Spectacled Eider (*Somateria fischeri*), a threatened Arctic sea duck.

A Bayesian approach is appropriate for classification decisions because Bayesian analyses yield probability distributions (called posterior distributions) that can be used directly in making decisions. For example, we may wish to classify a population as endangered if it is declining faster than 5% per year. However, when trend data are estimated (uncertain), we are not sure how fast the population is really declining. Despite the uncertainty, we still want to be confident that we are making correct decisions. We ask, therefore, "What is the probability that available trend data came from a population declining at a rate exceeding 5% per year?" This question cannot be answered directly by classical statistical inference. Classical inference, including hypothesis tests, P values, and confidence intervals, can be referred to as frequentist statistics because the theory is based on probabilities as long-run relative fre-

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[2] For reprints of this group of papers, see footnote 1, p. 1034.

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quencies (Iversen 1984). Unlike classical analysis, the direct probability statements generated from a Bayesian approach allow us to (1) more appropriately incorporate various types of uncertainty into our conclusions regarding the population growth rate, (2) easily and clearly perform a decision analysis to classify the population, and (3) easily perform an improved type of population viability analysis.

Ideally, classification decisions should be based on broad knowledge about a species' biology and vulnerability to anthropogenic and other threats. Population viability analysis (Gilpin and Soulé 1986) was developed to estimate extinction probabilities by integrating data on the many risks a species faces. For most species, however, data for quantifying risks are not only limited but also often contain a high degree of uncertainty (Taylor 1995, Ralls and Taylor, *in press*). This is not necessarily because the data are of poor quality, but may be because the species is rare. For example, as abundance declines, the precision of abundance estimates can also decline (Gerrodette 1987, Taylor and Gerrodette 1993). Therefore, decisions often must be based on few data that contain large uncertainties.

Our problem, then, is to present uncertain scientific information such that decision makers know their chances of committing classification errors. We define two types of errors: the overprotective error of classifying a species to a protection category that it does not warrant, and the underprotective error of classifying a species to a low-risk category or not classifying it at all, when it warrants greater protection. These errors are inversely related: the higher the probability of an overprotective error, the lower the probability of an underprotective error, and visa versa. Thus, the potential error of protecting a relatively secure species at the cost of alternative objectives, including protection for more vulnerable species, is inversely related to the potential error of failing to protect or recover truly vulnerable species. When making classification decisions, managers should be aware of the trade-offs between these errors.

Reaching a decision about species classification takes several steps. We must select: (1) a data set, (2) criteria for species classification using those types of data, and (3) decision criteria in the form of loss functions or acceptable trade-offs of classification errors. We then (4) analyze the data to estimate losses for decisions or estimate the error probabilities. In this paper, we want to compare decision making using Bayesian and frequentist approaches. To reduce complexity, we illustrate the comparison using only one type of classification decision as an example: changing classification from threatened to endangered status.

The best available data for assessing extinction risks for the Spectacled Eider are abundance index estimates through time. Although the recovery team developed alternative classification criteria based upon absolute abundance estimates, present surveys are designed only to detect trends in abundance indices and do not provide reliable data on absolute abundance.

The second step is choosing classification criteria. These choices define "endangered" and "threatened" classifications in quantitative terms. Although classification decisions must be based upon scientific evidence and are often supplemented with population modeling, the threshold criteria for placing species in specific categories are not dictated by science. Classification criteria reflect levels of species extinction risk and protection judged acceptable to managers and the public.

The Spectacled Eider recovery team selected a population declining by ≥5% per year as the trend criterion for endangered classification (U.S. Fish and Wildlife Service 1994). Based on population simulations, a growth rate of $r = -0.05$ indicated a 5% chance of reaching a critically small population of 250 adults within 50 yr. The simulation incorporated variability in the population growth rate. Population growth rates (r) vary temporally because of environmental stochasticity (fluctuations in the organism's biotic and abiotic environment that cause population-wide variation in birth and death rates). Although there are no data on variance in r for Spectacled Eiders, we do have estimates of the variance in r from two studies of Common Eiders (*Somateria m. mollissima*). Sampling error does not confound variability in r in these studies, in which nearly complete census data were available for 10 yr. A nonmigratory Common Eider population provides a likely estimate for minimum potential variance in r ($n = 10$ yr, standard deviation (s) = 0.07, Swennen 1983). A second population, which experienced a large die-off of adults in an oil spill, provides a likely estimate for maximum variability in r ($n = 10$ yr, $s = 0.21$; Milne 1974). Thus, the simulations the recovery team used to choose Spectacled Eider classification criteria drew s from a uniform distribution between 0.07 and 0.21. Thus, r' (for each year) was randomly chosen from a uniform distribution with mean = r, and s^2 = random number between 0.07 and 0.21 [symbolized U(0.07, 0.21)].

The third step in species classification is choosing decision criteria. Managers need to know the probability of making over- or underprotective errors, given a particular data set (for Spectacled Eiders, the time series of abundance index estimates). As with classification criteria, under- and overprotective error thresholds must be stated in quantitative terms. The Spectacled Eider recovery team considered as underprotected those populations declining at ≥5% per year that were not classified as endangered, and as overprotected those populations that were maintaining at least a 0% per year growth rate but were classified as endangered. They also preferred to have equal probabilities for the two error types. Further explanation of these decision criteria is presented in the draft Spectacled Eider Recovery Plan (U.S. Fish and Wildlife Service 1994).

Bayesian decision analyses use "loss functions" that explicitly assign a loss value to decisions (Berger 1985). For example, not classifying a population declining at 10% per year should be a greater loss than not classifying a population declining at 5% per year. Similarly, incorrectly classifying a population that was actually increasing by 1% per year should be a greater loss than incorrectly classifying a population decreasing by 1% per year. We give two examples of loss functions that are consistent with the recovery team's preference for equal probabilities of the error types. Alternatively, the decision could be approached in a more frequentist fashion. We compare the Bayesian loss function approach to a frequentist approach that calculates the probability of committing different errors, similar to a power analysis.

Finally, the data are analyzed to estimate the probability that the data resulted from differing hypotheses about growth rates. We detail a new technique, using Bayesian inference, that analyzes time series of abundance estimates to estimate a probability distribution for the population growth rate (in Bayesian terms, the posterior distribution). This Bayesian trend analysis explicitly incorporates data uncertainty into estimating probabilities for various population growth rates. A Bayesian decision analysis is performed by integrating the product of this probability distribution with the specified loss function to calculate what is called the "Bayesian expected loss," and then by choosing the action that minimizes this loss (Berger 1985). Therefore, the classification decision is made by choosing the classification that has the smaller Bayesian expected loss. This completes the classification decision process using Bayesian methods.

We also introduce a second method for making a classification decision that is based on the framework of a hypothesis test in classical frequentist statistics. It uses a simulation study to select a critical value of the estimated parameter (i.e., r) that will equalize over- and underprotection errors. This is essentially a frequentist power analysis that requires specific null and alternative hypotheses to estimate Type I and Type II errors (Cohen 1988). Fortuitously, because we have imbedded the Bayesian inference analysis inside this study, we can also use the simulation study to examine the classification error rate that results from using the specified loss function in the Bayesian decision analysis.

To demonstrate the full utility of the Bayesian trend analysis introduced here, we also perform a population viability analysis (PVA) using the posterior distribution that resulted from data analysis. This provides a new method of performing a PVA that can more fully incorporate various types of uncertainty into the calculation of an extinction probability distribution, particularly uncertainty in estimating the population growth rate from time series of abundance estimates.

Background: why use a Bayesian approach?

The primary reason for our using a Bayesian paradigm (Howson and Urbach 1991) is that frequentist hypothesis testing does not give us the answer to the question we want to ask: what are the relative probabilities of the competing hypotheses? We will not reiterate a full critique of frequentist hypothesis testing here (see Ellison 1996 for pointers to much of the literature), but two of the main critiques can be summarized as follows. First, any null hypothesis can be rejected (and similarly any significance test can be made significant) by choosing an appropriately large sample size (Press 1989). Thus, finding a significant difference does not make any statement about the magnitude of the difference, something that is usually of great importance to ecologists. This ambiguity is unsatisfactory and exposes the subjectivity behind claims of significance in classical statistics (a criticism that is often directed at Bayesian statistics but is rarely recognized in the frequentist approach). For example, with Spectacled Eiders, we are much more concerned with recognizing a large decline (>5% per year) in the population than we are in recognizing a statistically significant but small decline (e.g., <1% per year).

The ambiguity in classical hypothesis testing can be remedied somewhat by performing a power analysis to examine Type II errors that explicitly addresses the ability of the analysis to identify a decline of a specified magnitude (Gerrodette 1987, Taylor and Gerrodette 1993). However, power analyses can be difficult and are rarely performed in ecological studies (Ellison 1996). More importantly, Type I and Type II errors do not necessarily correspond closely to the probabilities of the hypotheses (Lee 1989). Classical statistics textbooks warn us that the statement that a null hypothesis is rejected at the 5% level does not mean that there is only a 5% chance that the hypothesis is true (Lee 1989), although many people intuitively or habitually form that conclusion. We want to be able to state the probability that the population is declining by >5% per year vs. the probability that the population is stable or increasing. The only way to make such a statement is to use Bayesian statistics. The ability to make such clear probabilistic statements about what the data tell us should improve decision makers' and the public's understanding of scientific results (Ellison 1996).

Bayesian methods also yield results that can be used directly in decision theory to make the classification decision. Any attempt to manage natural resources involves making decisions; therefore, decision analyses should be a part of ecological management. Forcing decision making into a hypothesis testing framework can lead to problems. For example, the loss from making an incorrect decision will often increase as a function of the distance of the estimated parameter from the true situation (which is probably the case here, since the risk of extinction is greater the more rapidly Spec-

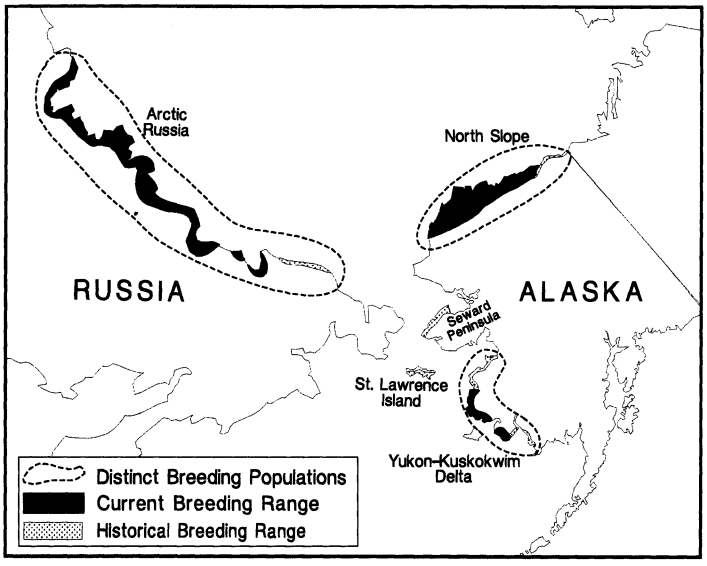

FIG. 1. Breeding distribution of Spectacled Eiders. Historical breeding range (shaded) is based on anecdotal information. Current breeding range (solid black) indicates recent confirmation of presence during surveys. (Figure by P. Gallagher.)

tacled Eiders are declining), and this loss cannot be correctly measured by classical error probabilities (Berger 1985). Although hypothesis testing is considered a part of inference theory, not decision theory, its form of construction (hypotheses are rejected or not) lends itself to making decisions. Barnett (1982) indicates that a decision theory analysis can be constructed that equates to the standard hypothesis test; however, it would be based on a principle that seems to have limited appeal in decision theory terms. However, performing a hypothesis test with Type I error set equal to Type II error should correspond to a decision analysis using a simple loss function. We will thus explore using just such a non-Bayesian decision criterion, by way of a Monte Carlo simulation technique that can be applied to any statistic regarding the population growth rate.

Finally, another reason for using Bayesian analysis methods in this case is the ability to fully incorporate various kinds of uncertainty into a single posterior probability distribution for the population rate of change. Taylor (1995) criticized PVA calculations for not accounting for uncertainty due to parameter estimation. Here, we present an example of a PVA for Spectacled Eiders that accounts for parameter estimation uncertainty.

Background: Spectacled Eiders

Spectacled Eiders are large-bodied sea ducks that nest in eastern arctic Russia and along the Chukchi and Bering Sea coasts of Alaska (Dau and Kistchinski 1977) (Fig. 1). Until recently, their marine range, which they occupy for most of the year, was unknown. Ongoing satellite telemetry tracking and aerial surveys have located some molting and wintering concentrations in the Bering Strait and northern Bering Sea region. Adapted to severe arctic and subarctic environments, eiders exhibit deferred sexual maturity, low annual recruitment rates to breeding age, variable annual rates of nonbreeding by adults, and high annual adult survival rates compared to more temperate duck species (Goudie et al. 1994). Few data relating to these life history traits are specific to Spectacled Eiders.

In 1993, the U.S. Fish and Wildlife Service listed the Spectacled Eider as a threatened species, due to a dramatic population decline in western Alaska (U.S. Fish and Wildlife Service 1993). The number of pairs nesting on the Yukon–Kuskokwim Delta (YKD), then thought to be the species' primary nesting range, had declined from an estimated 50 000 pairs in the early 1970s to <2000 pairs by 1992 (Stehn et al. 1993). Recent and long-term aerial survey data corroborated the declining trend. No quantitative data were available for examining population trends on Alaska's North Slope or in arctic Russia. Possible threats to the Alaska population include altered marine food supplies, contamination with toxic metals, lead poisoning from ingesting waste lead shot, increased predation, disease, and overharvest, but the causes of the long-term population decline are not known (Stehn et al. 1993, U.S. Fish and Wildlife Service 1994).

Background: abundance estimates for the Yukon–Kuskokwim Delta (YKD)

Trends in Spectacled Eider abundance are available only for the YKD, one of three primary breeding regions for this species. We use data from three independent surveys; two are based on eiders observed on aerial transects on the YKD coast, and the third is based on Spectacled Eider nests found on randomly located ground plots (Table 1). We use no estimates of absolute abundance, because the proportion of birds missed is unknown for any survey.

Since 1957, aerial transect lines of the North Amer-

TABLE 1. Abundance index (A.I.) and precision estimates for the three Spectacled Eider surveys of the YKD.

	Breeding Pair Survey		Ground Plots Survey		Aerial Coastal Survey	
Year	A.I.	CV	A.I.	CV	A.I.	CV
1957	25105	0.784				
1958	14472	0.744				
1959	16392	0.847				
1960	16983	0.714				
1961	14472	0.891				
1962	11371	0.869				
1963	10633	0.977				
1964	25548	0.878				
1965	23924	0.858				
1966	18607	0.794				
1967	15506	0.703				
1968	11224	0.603				
1969	12405	0.798				
1970	9747	0.723				
1971	7827	0.63				
1972	5316	0.607				
1973	9894	0.611				
1974	8270	0.65				
1975	7679	0.739				
1976	7089	0.766				
1977	8122	0.54				
1978	5021	0.459				
1979	8861	0.602				
1980	5464	0.803				
1981	6202	0.825				
1982	7089	0.799				
1983	3544	0.893				
1984	5612	0.811				
1985	3840	0.672				
1986	4135	0.573	6990	0.284		
1987	1477	0.791	5513	0.191		
1988	8270	0.726	5273	0.305	2785	0.148
1989	2658	0.859	5563	0.232	2260	0.173
1990	1477	0.977	4473	0.205	2400	0.155
1991	2363	0.977	2360	0.241	1735	0.165
1992	2658	0.859	1943	0.215	1998	0.168
1993	1477	0.781	1777	0.229	3185	0.126
1994	2363	0.496	2559	0.17	3290	0.124
1995	2363	0.496	2968	0.199	3981	0.139

Notes: The factor used to expand the abundance index for the Ground Plot Survey to the entire YKD coast is based on data from the Coastal Survey. For 1986 and 1987 (previous to the Coastal Survey), we used the mean expansion factor from the aerial surveys flown in 1988 and 1989.

ican Waterfowl Breeding Pair Survey have been flown in Alaska with a standard protocol (U.S. Fish and Wildlife Service 1987, Conant and Groves 1994). Flying at 30–45 m altitude and 72–82 m/s (140–160 knots), a pilot and an observer record all waterfowl species seen along 400 m wide strip transects. Portions of seven transects sampling 89.8 km^2 are located within the coastal portion (13 266.3 km^2) of the YKD, which includes the western Alaska nesting range of Spectacled Eiders.

Beginning in 1985, a more intensive aerial transect survey (the Coastal Survey) was flown over 12 852 km^2 of the coastal YKD (Butler et al. 1987, 1995). This survey was designed to monitor declining goose populations (Raveling 1984, King and Derksen 1986) and used a stratified design based on habitat features. Beginning in 1988, a third observer in the aircraft's rear seat recorded eiders, other ducks, gulls, and loons. Special emphasis was not put on counting eiders until 1990.

For both aerial surveys, the indicated total number of eiders is calculated as twice the number of single males observed plus all birds seen in pairs and flocks. Single males are easily observed whereas females are not; single males are assumed to indicate a breeding pair. An abundance index is estimated by multiplying the mean density (the indicated total birds divided by the transect area sampled) by the total area. Sampling error variance is calculated from the ratio of observed eiders per sampled area on each transect segment (Caughley 1977). Males start to leave the breeding grounds shortly after nest initiation. Variability in the timing of this event each year may lead to a variable proportion of males being available for counting.

These aerial surveys could bias trend estimates positively (underestimate the decline) for two reasons. First, because the accuracy of species identification has varied among years and observers, aerial survey data combine Spectacled, Common, and unidentified eiders. Based on nests found during recent years, 78% of all eiders on the central YKD are Spectacled Eiders (Stehn et al. 1994: Table 4). If Common Eiders have maintained constant abundance, they have become an increasingly large proportion of the total eider observations as Spectacled Eiders have declined, contributing a positive bias to the "eider" trend. Second, it is also likely that, during the early years of the Breeding Pair Survey when waterfowl populations were much larger than in recent years and eiders received little management attention, eiders were undercounted due to observer saturation. We cannot estimate this bias, but we know it would influence classification decisions only if the most likely value for population growth were slightly larger than the classification criterion ($r = -0.05$). This turns out not to be the case.

The Ground Plot Survey began in 1986 and sampled, with randomly located plots, some of the strata sampled by the Coastal Survey (Stehn et al. 1994). Each plot (1986–1994: 0.323 km^2; 1995: 0.451 km^2) was centered around a random x,y coordinate that was newly selected each year. Ground plots were searched for nests during a single visit of 2–6 h by two biologists. Nearly all nests were visible and could be identified to species by down and contour feathers found in the nest bowl, even if the adult birds were absent. Although the area and strata sampled on the ground changed somewhat through time, the aerial survey consistently covered the entire YKD coast. Therefore, the population estimates from ground samples were expanded to the total YKD coastal area, based on the inverse proportion of the aerially observed population falling within the ground sampled area to the entire aerial survey area. When calculating the expanded YKD nest estimate, we assumed that the ratio of Spectacled to Common Eiders was the same in all strata. In 1995, a subsample of plots

was searched a second time to quantify the proportion of missed nests using mark–recapture techniques. Similarly, paired observers on the coastal aerial survey were compared to estimate the proportion missed. We used preliminary analyses of these data, indicating the proportion of nests found in a single visit to be 0.73 and the proportion observed during aerial counts to be 0.74 (Bob Stehn, *personal communication*), to give corrected abundance estimates. We believe these abundance estimates to be close to the absolute abundance and, therefore, use them with equal weighting in the PVA.

Methods

Bayesian trend analysis

Our primary interest is in estimating the population growth rate from observed data, which are abundance index estimates with associated precision estimates. We assume an exponential model of population growth

$$N_{t+1} = N_t e^{(r+\epsilon_t)}, \tag{1}$$

where N is the number of breeding pairs, t is time (years), r is population growth rate, and ϵ_t is Gaussian distribution symbolized as $G(\bar{X}, s_r^2)$, where $\bar{X} = 0$. When $s_r = 0$, the model is deterministic. When $s_r > 0$, then the model is stochastic, with annual growth rate drawn from a distribution. The parameter of interest for classification decisions is r. Parameters are estimated by fitting the model with the available time series of abundance estimates using the same Bayesian analysis methods as Wade (1994). This is analogous to a weighted nonlinear regression using classical statistical methods (essentially because the contribution of each abundance estimate to the estimation is weighted by the likelihood, which is a function of its estimated precision).

Due to the nature of the available data set, several other parameters have to be defined and estimated. The abundance estimates and their associated coefficients of variation, CV (which were estimated from the sampling design for each survey), represent the observed data in this analysis. Because the number of eiders that breed and are thus available to be counted may vary through time (more than explained by trends in r), the estimated CV may not account for all of the variance in the abundance estimates. Therefore, we estimate a parameter (m) for the additional lack of fit to the model not accounted for by CV. This parameter multiplies the estimated CV to give a total estimated variance

$$s^2 = (m\,\widehat{\mathrm{CV}}(N)\hat{N})^2, \tag{2}$$

where s^2 = variance, and the circumflex indicates the estimated value from the survey data. A different parameter is specified for each of the three different abundance time series (m_B, m_G, m_C, where subscripts denote B for breeding, G for ground plot, and C for coastal surveys). The estimated CVs account for the precision of the surveys to estimate the number of eiders that are in the study area, whereas the purpose of the CV multipliers is to account for additional variance in the abundance estimates, such as the number of eiders that breed in each year and, thus, are available to be surveyed. These multipliers are assumed to be constant through time.

Two scaling parameters are required to scale the time series of abundance indices from different surveys to one another. This is because none of the survey estimates represents an estimate of absolute abundance, but instead serves as a relative index of abundance. Therefore, we have arbitrarily chosen one of the surveys (the Breeding Pair survey) as the default unit of measure of relative abundance; data from the other two surveys are scaled to these units. The two scaling parameters (a_G and a_C) are estimated as part of the overall analysis. In summary, we estimate as many as eight parameters in the analysis.

We use Bayes' theorem to estimate the specified parameters from the abundance data. This theorem states that the probability of parameter θ, given the data x, written $p(\theta \mid x)$, is proportional to the product of the probability of the data x, given parameter θ, written $p(x \mid \theta)$, and the probability of the parameter, $p(\theta)$, not conditioned upon the data x. The probability $p(\theta \mid x)$ is called the posterior distribution for parameter θ, and is the end result of the analysis. The probability $p(x \mid \theta)$ is called the likelihood function, and is often written as $L(\theta \mid x)$. The probability $p(\theta)$ is called the prior distribution for θ, and represents the probability distribution for θ before the data x were known. Thus, the posterior distribution is equal to the product of the likelihood function and the prior distribution, normalized by the integral of the product of the likelihood and the prior:

$$p(\theta \mid x) = \frac{L(\theta \mid x)p(\theta)}{\int L(\theta \mid x)p(\theta)\,d\theta}. \tag{3}$$

In our analysis of the Spectacled Eider, we have specified more than one parameter; here, θ represents up to eight parameters. Thus, the integration in the denominator, which calculates the normalizing constant, is actually multidimensional, with dimension equal to the number of parameters being estimated.

The likelihood function for the parameters in a population model, given a time series of abundance estimates, is relatively easy to write and calculate (e.g., De la Mare 1986). In any single year, the likelihood of an observed abundance estimate in year t, $N(t)$, given a specified model population size in year t, N_t, is straightfoward: it is the likelihood function defined by the assumed sampling distribution of the abundance estimate. Here, we assume the sampling distribution of the abundance estimates is distributed as a Gaussian

distribution with estimated mean $N(t)$ and standard deviation $S(t)$. Thus, the likelihood is:

$$L(N_t \mid N(t), S(t)) = \frac{1}{\sqrt{2\pi}\, S(t)} e^{-(1/2S(t))(N(t)-N_t)^2}. \quad (4)$$

Although N_t is not an explicit parameter of the model, the model parameters N_0 and r uniquely determine a population trajectory $N_1, N_2, \ldots N_c$ (where c is the last year projected to, the current year). Therefore, the total likelihood for N_0 and r, given the data, is the product series of all the individual likelihoods of the N_ts (the model trajectory), given the $N(t)$s (the time series of abundance estimates).

More formally, the likelihood for the three surveys combined was:

$$\begin{aligned} &L(\theta \mid \overline{N_B}, \overline{N_G}, \overline{N_C}) \\ &\equiv p(\overline{N_B}, \overline{N_G}, \overline{N_C} \mid \theta) && (5)\\ &= p(\overline{N_B} \mid \theta)\, p(\overline{N_G} \mid \theta)\, p(\overline{N_C} \mid \theta) && (6)\\ &= \prod_{t=1}^{n} p(N_B(t) \mid \theta)\, p(N_G(t) \mid \theta)\, p(N_C(t) \mid \theta) && (7)\\ &= \prod_{t=1}^{n} \big[L(N_0, r, m_B \mid N_B(t), \mathrm{CV}(N_B(t))) \\ &\qquad \times L(N_0, r, m_G, a_G \mid N_G(t), \mathrm{CV}(N_G(t))) \\ &\qquad \times L(N_0, r, m_C, a_C \mid N_C(t), \mathrm{CV}(N_C(t)))\big] && (8)\\ &= \prod_{t=1}^{n} \Big[\frac{1}{\sqrt{2\pi} s_B(t)} e^{-(1/2)[(N_B(t)-N_t)/(s_B(t))]^2} \\ &\qquad \times \frac{1}{\sqrt{2\pi} s_G(t)} e^{-(1/2)[(N_G(t)a_G-N_t)/(s_G(t))]^2} \\ &\qquad \times \frac{1}{\sqrt{2\pi} s_C(t)} e^{-(1/2)[(N_C(t)a_C-N_t)/(s_C(t))]^2}\Big], && (9) \end{aligned}$$

where N_t is the model population size in year t, as projected from N_0 and r using Eq. 1; $N_I(t)$ is the abundance index estimate for Survey I in year t; a_I is the scale parameter that scales an abundance estimate from survey I to the Breeding Pair survey; I is the index of the survey type, where B indicates the Breeding Pair surveys, G indicates the Ground Plot surveys, and C indicates the Coastal surveys; $S_I(t)$ is the total standard deviation of abundance estimate $N_I(t)$, and is defined as:

$$S_I(t) = m_I\, \widehat{\mathrm{CV}}\, (\hat{N}_I(t))\, \hat{N}_I(t), \quad (10)$$

where m_I is the CV multiplier for survey I, and $\mathrm{CV}(N_I(t))$ is the estimated survey precision (CV) of abundance estimate $N_I(t)$. The parameter values that maximize Eq. 4 are the maximum likelihood estimates, a common point estimator in frequentist statistics. In Bayesian statistics, rather than maximizing Eq. 4, we need to integrate the product of Eq. 4 and the prior distribution of the parameters.

Until fairly recently, most Bayesian analyses were restricted to cases in which the prior distribution could be chosen so that it was "conjugate" to the likelihood distribution, resulting in their product being a distribution of a known form, which was integratable by analytic methods. Advances in computing power and numerical and Monte Carlo integration methods have removed this restriction. We use the Sampling–Importance–Resampling routine of Rubin (1988), which Smith and Gelfand (1992) advocate as a particularly useful and simple integration technique for Bayesian statistics. In this method, values for the parameters are randomly selected from their joint prior distribution to form a sample set θ_i. The likelihood of the data, given this particular θ_i, is calculated and stored. This is repeated, generating n_1 θ_is with associated likelihoods. These n_1 θ_is are then resampled n_2 times with replacement, with probability equal to weight q_i, where

$$q_i = \frac{L(\theta_i \mid x)}{\sum_{j=1}^{n_1} L(\theta_j \mid x)}. \quad (11)$$

Rubin (1988) showed that this generates a random sample from the joint posterior distribution of size n_2. The resampling with replacement from the n_1 θ_is with weight q_i makes this process analogous to a weighted bootstrap procedure. We set values for n_1 and n_2 to yield smooth posterior distributions, which depended on the number of parameters estimated. For single survey analyses, $n_1 = 300\,000$ and $n_2 = 5000$. For analyses of all three surveys, $n_1 = 8\,000\,000$ and $n_2 = 10\,000$. For analyses of all surveys with environmental stochasticity (Eq. 1), $n_1 = 25\,000\,000$ and $n_2 = 10\,000$. With the exception of initial population size, prior distributions were uniform distributions that were iteratively set to encompass the values of the posterior distribution for each parameter.

For the full analysis using the deterministic population model (Eq. 1) and all three data sets, values were randomly drawn from their prior distributions for the seven parameters (r, N_0, m_B, m_G, m_C, a_G, and a_C) to form a θ_i. Those parameter values were then used to project a model population trajectory using Eq. 1, forming a series of model population sizes $N_1, N_2, \ldots N_c$. The likelihood of that population trajectory, given the data, was then calculated using Eq. 9, and the θ_is and associated likelihood value were stored. This was repeated n_1 times to form the initial sample. Then the θ_i were resampled n_2 times with replacement with weight q_i (Eq. 11) to form the second sample, which represents a random sample from the posterior distribution and is thus used to approximate the posterior distribution (Fig. 2).

We analyze the three time series both together and separately to illustrate the effects of different quantities and qualities of data on the classification process. The seven parameters necessary for the analysis of all three

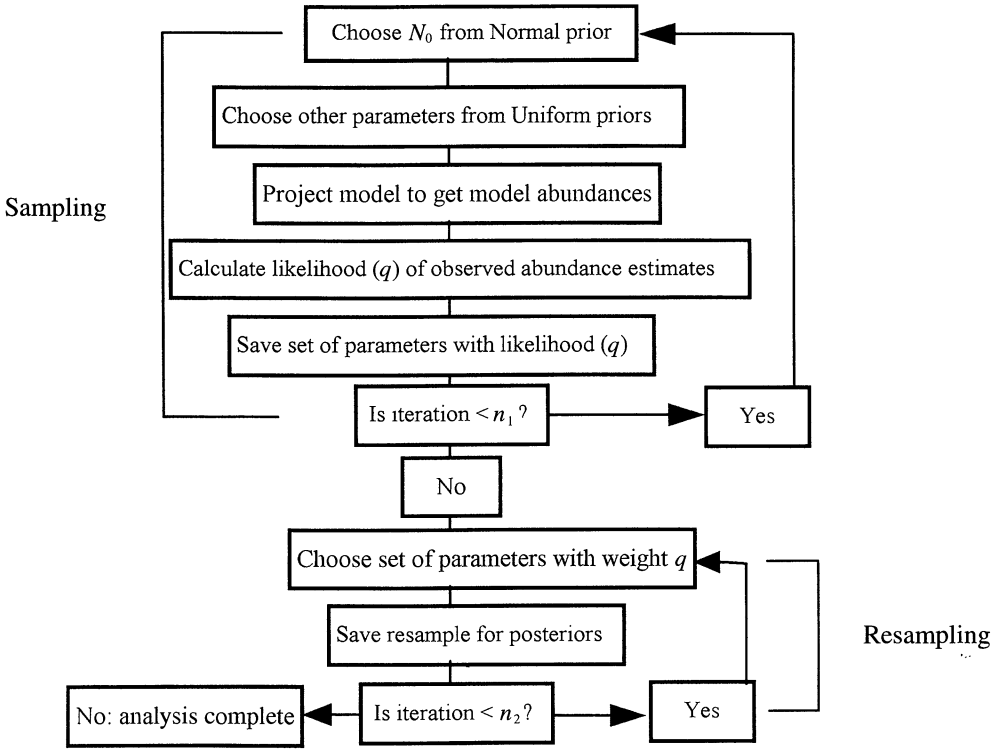

FIG. 2. Schematic of the sampling–resampling algorithm used to conduct the Bayesian analysis.

data sets were given prior distributions as follows: (1) N_0 (initial population size in 1957, in units of Breeding Pair Survey) $G(23\,165,\ 19\,111^2)$; (2) r [uniform U(−0.095, −0.045); (3) m_B, multiplier for the variance in Breeding Pair Survey U(0.3, 0.8); (4) m_G, multiplier for the variance in Ground Plot Survey U(0.7, 3.5); (5) m_C, multiplier for the variance in Coastal Survey U(0.8, 7.5); (6) a_G, scaler of abundance index for Ground Plot to Breeding Pair Survey U(0.35, 2.90); (7) a_C, scaler of Coastal to Breeding Pair Survey U(0.35, 2.50). The prior for N_0 is based on the first abundance estimate from the Breeding Pair survey in 1957. The other priors were chosen to be uniform distributions to represent no prior knowledge of their value. For practical purposes, we bounded the prior distributions because extreme values had nearly zero likelihood. We set these bounds after a few trial runs to ensure that the priors included all possible values of the posterior distribution.

The classification criterion for endangered status set by the recovery team ($r \leq -0.05$) was based on simulations that included environmental stochasticity. Environmental stochasticity makes the long-term growth rate less than the expected growth rate (r). For example, a population with mean $r = 0.00$ and some amount of variance will not fluctuate around the initial abundance, but rather will decline. Expected growth rate and long-term growth rate differ because the latter is actually the geometric mean of the distribution of growth rates, which is always less than the arithmetic mean. To check that our classification decisions are not influenced by the omission of environmental stochasticity, we reanalyzed the full data set using the stochastic population model (Eq. 1). The prior distribution for the additional parameter s_r, the standard deviation of the growth rate, was specified as follows. We used both a uniform distribution, as was done by the recovery team, and a worst case scenario using only the highest value for variability in r. These values were based on the data available for Common Eiders. Therefore, the prior for s_r was either U(0.07, 0.21) or was fixed at 0.21.

This prior distribution for s_r points out another unique advantage of Bayesian methods: an explicit framework for incorporating prior knowledge into an analysis. It is well known that, when fitting a population model to abundance data, it is impossible to distinguish between environmental variance and sampling error from the abundance data alone (Hilborn and Walters 1992). This has led researchers to ignore one or the other. Use of a Bayesian framework has enabled us to incorporate both, because the sampling error is accounted for by the CVs and their multipliers, and uncertainty due to environmental variance is incorporated into the prior distribution for s_r, using data from Common Eiders. Admittedly, the data on environmental variance are not ideal, but they are preferable to either ignoring stochastic population dynamics or assuming a noninformative prior that might overemphasize the importance of variance in r.

TABLE 2. Loss functions for the decisions to classify or not classify a population as endangered. A hypothetical probability that r (population growth rate) was in the various states is given in the last row to illustrate the example in the text.

Decisions	Loss functions		
	$r \leq -0.05$	$-0.05 < r < 0$	$r \geq 0$
Classify as endangered	0	0	1
Do not classify as endangered	1	0	0
Probability of event	0.3	0.5	0.2

Bayesian decision analysis

We demonstrate two types of loss functions that are consistent with the recovery team decision criterion of equalizing overprotection (classifying a population with $r \geq 0.00$) and underprotection (not classifying as endangered a population with $r \leq -0.05$) errors. The first type are very simple loss functions. The simple underprotective loss function assigns a loss of one to the incorrect decision not to classify and a zero loss to any other outcome. The simple overprotective loss function assigns a loss of one to the incorrect decision to classify and a zero loss to other outcomes (Table 2). The probability of underprotection is the portion of the posterior distribution for r that is ≤ -0.05. The equivalent for overprotection is the portion of the distribution $r \geq 0.00$. Thus, if the proportion of the posterior distribution for r that is ≤ -0.05 is greater than the proportion ≥ 0.00, then the loss is minimized by an endangered classification. If the opposite is true, then not classifying as endangered will minimize loss. An example is given in Table 2, where the underprotective loss is $1 \times 0.3 = 0.3$ and the overprotective loss is $1 \times 0.2 = 0.2$. Therefore, to minimize loss, the decision would be to classify the population (thereby avoiding the higher loss of underprotecting the population).

This simple loss function gives an equal loss to not classifying a population declining at 5% per year and a population declining at 10% per year. Clearly, the latter population is at higher risk and should warrant assignment of a higher loss value. Thus, our second type of loss function quantifies this risk, using the probability that a population reaches a critical level (defined here as 250 adults) within a specified time. We simulated population trajectories as follows. For rates of decline from $r = 0.0$ to $r = -0.1$, (1) choose N_0 from the 1995 estimate for the ground plot survey; (2) choose s_r from U(0.07, 0.21); (3) choose r' from $G(r,s_r^2)$ for each year; (4) project population until $N < 250$; and (5) repeat steps 1–4 10 000 times for each r, saving the year each population falls below 250 adults. The resulting loss function can be seen in Fig. 3. To represent the decision to equalize errors, we set the loss function for overprotection to reflect the underprotection function with a zero loss occurring when $r = -0.05$ (no loss because it is a correct decision). Calculation of loss for each error type is done by multiplying the chosen posterior distribution for r by each loss function.

Hypothesis test-based decision

We also demonstrate a frequentist analysis using the posterior distributions generated from the Bayesian trend analysis. As in a typical power analysis, we estimate errors using distributions for the hypotheses under consideration, namely $r = 0.00$ or $r = -0.05$. We use the proportion of the posterior distribution ≤ -0.05 as our test statistic. We then need to generate a statistical table that gives the probability of over- and underprotective errors for different values of the test statistic. To do this, we run simulations assuming $r = 0.00$ to estimate the overprotective error. A similar set of simulations is run to estimate the underprotective error ($r = -0.05$).

For each simulation, the following protocol is followed: (1) randomly choose N_0 from a Gaussian distribution with mean $= N_0$, estimated from the data, and s^2 according to Eq. 2, drawing from the posterior distribution for m; (2) project model abundance through the length of the time series; (3) generate simulated abundance estimates by using the estimated CV from the data for the appropriate year and choosing other parameters from their posterior distributions (m for each survey and scaling parameters); (4) follow this protocol for generating a posterior distribution from trend data, but use the simulated estimated abundances rather than the observed estimated abundances; (5) save the proportion of the posterior distribution ≤ -0.05 and the proportion ≥ 0.00; (6) repeat steps 1–5 500 times. The error probability is estimated as the proportion of times the simulations were incorrectly classified for a given critical value. For example, consider four simulations where $r = 0.00$ that resulted in the following

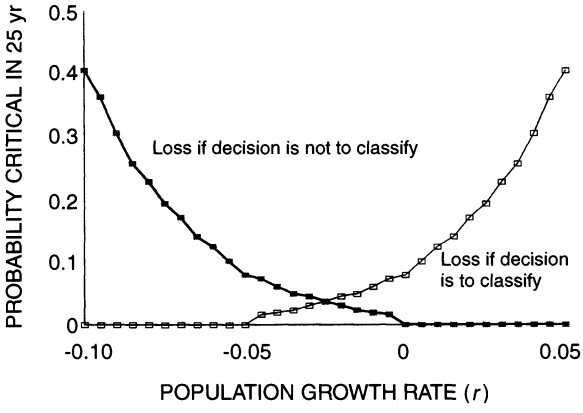

FIG. 3. Loss functions using the proportion of populations expected to reach critical numbers of 250 adults in 25 yr as a measure of risk. The overprotective loss function (open squares) is set to yield equal probabilities of over- and underprotective errors. Because classification decisions minimize loss, the actual ordinate scale is irrelevant as long as both loss functions are in the same relative units.

TABLE 3. Statistics for the posterior distribution for r, population growth rate.

Survey	Mean r	95% confidence limits Lower	Upper
Breeding Pair	−0.066	−0.076	−0.056
Breeding Pair (s_r = 0.21)	−0.064	−0.122	−0.007
Ground Plot	−0.158	−0.267	−0.048
Coastal	+0.079	−0.061	+0.219
Combined (s_r = 0.00)	−0.067	−0.077	−0.057
Combined (s_r: U(0.07, 0.021))	−0.064	−0.104	−0.024
Combined (s_r = 0.021)	−0.041	−0.101	−0.019

proportions of the posterior ≤−0.05: 0.01, 0.06, 0.12, 0.17. A critical value of 0.10 would result in a 50% overprotective error, whereas a critical value of 0.15 would result in a 25% overprotective error. A critical value is sought that equalizes the two errors. If the observed value (proportion of the posterior distribution ≤−0.05) exceeded the critical value, then the population would be classified as endangered.

Bayesian population viability analysis (PVA)

The resampled output of the Sampling–Importance–Resampling (SIR) algorithm is 10 000 (n_2) sets of values for the eight parameters, which represents the joint posterior distribution. We utilize this joint posterior distribution to perform a PVA that includes the uncertainty in our data (the abundance index surveys) and the model (prior information on variability in r). We sequentially used each set of parameter values as input parameters for a stochastic population projection. Required input parameters are: N_0, r, and s_r. Each simulation began with N_0 individuals. The initial number, N_0, was drawn from either the 1995 estimated abundance distribution for the Coastal or the Ground survey with equal probability. The simulation proceeded according to Eq. 3, where r' was drawn from $G(r,s_r^2)$. Both the time to critical status (<250 adults) and the time to extinction (<2 adults) were stored.

RESULTS

Trends in abundance for the YKD

Table 3 gives descriptive statistics for the posterior distributions for r (all matching Gaussian distributions) for the three surveys separately and for the combined surveys. Also given are selected simulations where environmental stochasticity was included. Inclusion of variability in r altered the mean slightly and increased the spread of the posterior. We analyzed both a worst case scenario with the highest variability expected (s = 0.21) and a scenario that matched the recovery team simulations for classification criteria [s_r from U(0.07, 0.21)]. Only the use of extreme variability (resulting from a mass die-off of Common Eiders in an oil spill) resulted in an increase of r of >2%. As expected from visual inspection of the trend data (Figs. 4a and b), the Coastal Survey yields the least precise posterior distribution, whereas the longer Breeding Pair Survey yields the most precise distribution. When the data are combined into a single likelihood calculation (with scaled abundance indices), the mean estimate for r = −0.067 and 99.86% of the distribution is ≤−0.05 (Fig. 5). This composite posterior distribution closely resembles the posterior distribution for the Breeding Pair Survey.

The multiplier used to adjust the estimated sampling error to match the total variance for the time series (m_B, m_G, m_C) indicates that, although the Breeding Pair Survey underestimated precision, the other surveys overestimated precision (Table 4). The resulting posterior distributions for the multipliers were not influenced by the ranges chosen for the prior distributions. The ranges for the prior distributions for the multipliers were set iteratively by beginning with large ranges and reducing

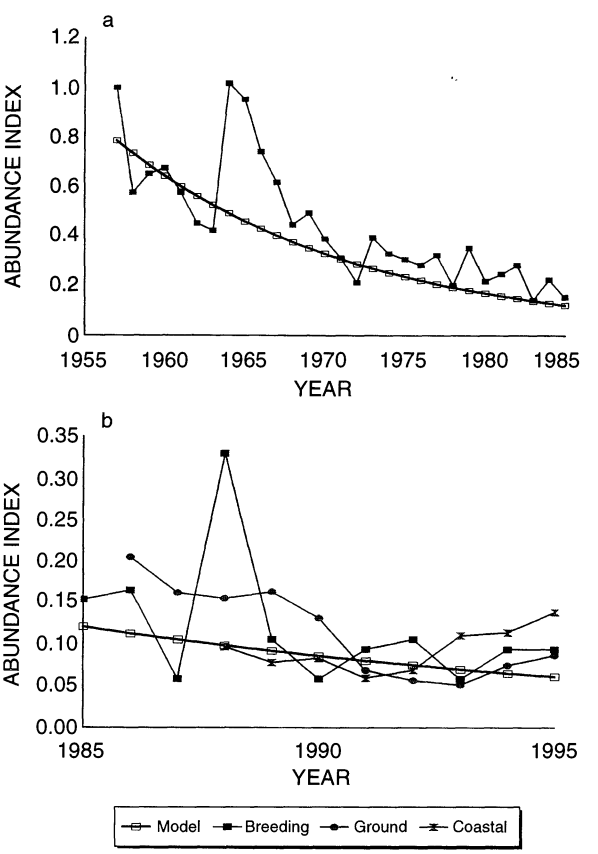

FIG. 4. Abundance indices and model estimates from 1957 to 1985 (a) and from 1985 to 1995 (b), where all trajectories have been scaled so that the abundance index for the Breeding Pair Survey in 1957 was 1.0. Two figures are shown to allow more detail in the most recent decade (note different ordinate axes). The Ground Plot Survey and the Coastal Survey have also been scaled to the Breeding Pair Survey using the mean posterior distribution estimate for the scaling parameters (a_G = 1.36 and a_C = 1.15, respectively). The model trajectory projects the mean posterior distribution values for all eight parameters.

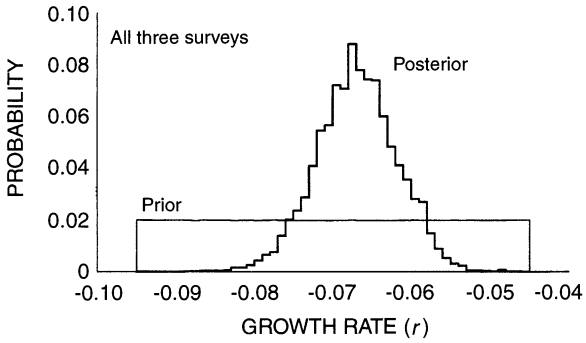

FIG. 5. Prior and posterior distributions for r for the full eight-parameter model, where s_r is drawn from U (0.07,0.21).

the range to include only the values with some likelihood and therefore some probability in the posterior distribution.

Decision analysis

Use of the simple zero–one Bayesian loss functions results in an endangered classification for all data sets except the Coastal Survey (Table 5). Introduction of environmental stochasticity would in no case influence the classification decision. Use of the more complex loss functions (Fig. 3) results in the same classification decisions in this case. For example, multiplying the loss functions in Fig. 3 by the posterior distribution in Fig. 5 results in an underprotection loss of 0.119 compared to an overprotection loss of 0.009.

Results for the frequentist analysis are essentially the same: classify in all cases except when the Coastal Survey is used alone (Table 6). Recall that this analysis requires generation of 500 posterior distributions each for both hypotheses ($r = 0.00$ and $r = -0.05$). For each one of these simulations, we stored the proportions of the distribution ≤ -0.05 and ≥ 0.00. This allowed a direct comparison of the performance of the frequentist technique to the Bayesian technique using a zero–one loss function. For the ground survey alone, the probabilities of over- and underprotective errors are equal at 0.82 (an 18% chance of committing either error). Using the zero–one loss function resulted in correct decisions when $r = 0.00$ with probability 0.78, and when $r = -0.05$ with probability 0.82. Therefore, for this simple case, results are very comparable. We could not equalize errors for the combined data sets or for the Breeding Pair Survey because the probability of committing an overprotection error was zero; thus, errors could not be equalized.

TABLE 4. Statistics for the posterior distribution for the standard deviation multipliers (m_i) for analysis of the individual surveys. Results for the combined surveys were very similar. We give percentiles of the posterior distributions because they are skewed right.

Survey	Percentiles		
	5th	50th	95th
Breeding Pair (m_B)	0.38	0.46	0.57
Ground Plots (m_G)	0.84	1.54	2.55
Coastal (m_C)	0.99	1.77	3.65

TABLE 5. Proportion of posterior distributions for r, with resultant Bayesian classification using a loss function that assigns a loss of 1 for an incorrect classification and zero otherwise.

Survey	Posterior mean	Proportions		Classify as endangered?
		≤ -0.05	≥ 0.00	
Breeding Pair	−0.066	0.999	0	Y
Breeding Pair ($s_r = 0.21$)	−0.064	0.689	0.014	Y
Ground Plot	−0.158	0.973	0.002	Y
Coastal	+0.079	0.036	0.864	N
Combined ($s_r = 0.00$)	−0.067	1	0	Y
Combined (s_r: U (0.07, 0.021))	−0.064	0.749	0.001	Y
Combined ($s_r = 0.021$)	−0.041	0.406	0.142	Y

PVA

Fig. 6 shows the cumulative "extinction" distributions for the states of critical ($N < 250$) and extinct ($N < 2$). Use of the joint posterior distributions made this a very simple and rapid procedure that includes uncertainty in parameter estimates as well as uncertainty about parameters for which we have no direct data (s_r). The breadth of these cumulative distributions is greater than those generated when parameter uncertainty is omitted. For example, simulations of Steller's sea lions using best estimates of parameters from 25 yr of data spanned only 60 yr (Taylor 1995), whereas the extinction distribution in Fig. 6, based on 38 yr of data, spans 528 yr, although 98% of the populations had gone extinct in a span of 211 yr.

DISCUSSION

Our Bayesian approach incorporated uncertainty in the data to estimate the probability of classification errors. The resulting posterior distribution of r allows both easy decision analyses and generation of a more realistic extinction distribution. Although the frequentist approach to decision analysis yielded the same results, the hypothesis-testing framework is very awkward, very difficult to explain, and computationally intensive. For example, the 1000 simulations required for a single frequentist decision analysis took ≈2 d on

TABLE 6. Error levels for the different surveys.

Survey	Underprotective error	Overprotective error	Critical value†	Observed value†	Classify as endangered?
Breeding					
Ground	0	0	<0.01	0.9988	Y
Coastal	0.18	0.18	0.21	0.9692	Y
Com-	0.2	0.54	0.26	0.0394	N
bined	0	0	<0.01	0.9986	Y

† Proportion < -0.05.

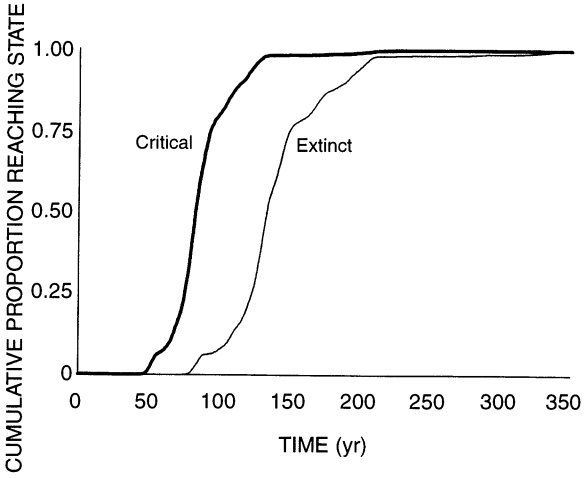

FIG. 6. Cumulative probability of populations reaching the states of critical ($N < 250$) and extinct ($N < 2$) with time. Distributions result from 10 000 simulations using the joint posterior distributions from the eight-parameter model.

a Pentium 90 MHz PC, whereas either Bayesian analysis was done in seconds. The use of loss functions also forces parties involved in the decision to specifically address the costs of errors. Loss functions also give greater flexibility than the hypothesis-testing framework because they allow consideration of a range of outcomes rather than only two (the null and alternative hypotheses).

Using extinction distributions that incorporate uncertainty is the only way that the risk to different species, or even different populations, can be compared with one another (Taylor 1995). Generation of risk distributions will be simple once absolute abundance estimates become available for Spectacled Eiders.

Only the analysis of the Coastal Survey by itself would not justify classification of YKD Spectacled Eiders as endangered under the recovery team's criterion. The posterior distribution of the variance multiplier (m_C) indicates that it is likely that the precision of this survey is overestimated. Nonetheless, neither method of decision analysis justifies an endangered classification based on this survey alone. The conflicting results of the Coastal and Ground Plot surveys are particularly curious in that they were conducted over about the same period. Even if population growth for the YKD Spectacled Eiders has changed (presumably increasing) over the past few years, one would not expect to get results from two simultaneous surveys strongly suggesting trends in opposite directions. Possibilities for the discrepancy are: (1) a differing proportion of males available during the aerial survey, contributing a greater than expected level of imprecision; (2) undercounting in the first few years of the aerial survey before eiders were emphasized; or (3) a lag time between the number of birds observed (which may include immature birds) and the number of nests observed. Clearly, more research is required to resolve this mystery.

The last three years of increasing trends for the Coastal and the Ground Plot survey highlight the need for further research on model choice. Clearly, the 38-yr decline in the Breeding Pair Survey has a very high influence on the posterior distribution for r. If population recovery occurs, meeting delisting criteria will be difficult, using all time series data and assuming an exponential model. Modelers could choose either to use a more complex model that allows both declines and increases, or to truncate the time series and keep the simple exponential model. Although the recovery team's reclassification criteria from endangered to threatened require use of ≥ 10 yr of data, they do not explicitly discuss how to analyze a time series of a once-declining, then recovering population.

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